B Trials per cell

Table 8 gives trials per cell (one cell = task, model, length, density, and alphabet or modulus value) for the data pooled in Figures 2, 4, and 5. Gemini uses 3.

Table 8: Trials per cell, $N \leq 64$. Q2.5: Qwen2.5-Instruct; Q3.8: Qwen3.8-27B; Gemma 4: Gemma-4-31B-it.

Task	Q2.5-7B	Q2.5-14B	Q2.5-32B	Mistral	Phi	Q3.8	Gemma 4
`cyclic_shift`	20	20	20	40	20	20	20
`random_permutation`	40	40	40	40	20	20	20
`conditional_shift`	40	20	40	40	20	20	20
`copy_previous`	40	40	80	40	40	40	40
`running_xor`	40	40	40	40	20	40	40
`running_sum_local`	40	40	80	40	20	40	40
`sliding_window_max`	20	20	40	20	20	20	20
Reset variants	80	–	20	–	–	20	20
Spaced versions (each)	20	20	20	20	20	20	20
Digit versions (each)	20	20	20	20	20	20	20

Checkpoints. Hugging Face revisions: Qwen2.5-7B-Instruct `a09a35458c70`, Qwen2.5-14B-Instruct `cf98f3b3bbb4`, Qwen2.5-32B-Instruct `5ede1c97bbab`, Mistral-7B-Instruct-v0.3 `c170c708c41d`, Phi-3.5-mini-instruct `2fe192450127`, Qwen3.8-27B `37561816e66b`, Gemma-4-31B-it `842da3794eaa`.

Generation budget. The no-marker run and every spacing $k \geq 2$ share one budget, $1.5N+20$ new tokens; $k = 1$ uses $2N + 20$. To check that the budget does not drive any effect, we re-tokenized every stored raw output of the seven open models ($N \leq 64$; 86,400 generations, since older letter-task files store only the clipped answer) and counted those that used the whole budget while giving fewer than N answer characters on the first line. This is an upper bound on cut-off answers, because it also counts short answers followed by unrequested text. The share is 0.58% without markers, 0.49% at $k \geq 2$, and 0.31% at $k = 1$; at the four stronger models it is 0.04%, 0.02%, and 0%. Nearly all cases are Phi-3.5-mini on spaced letters (up to 19% of one cell without markers, against 14% with), and on the dependency tasks the share is 0.03% or less in every class.

C How the uncertainty was computed

Pooled effects use the length-matched estimator of Section 3, with analytic standard errors over generations. Two checks ask how much that choice matters (`scripts/analyze_statistics_robustness.py`; per-cell values in `results_clean/_summary/statistics_robustness.csv`). Both resample 2,000 times, over the 91 pooled task–model cells of the three position-local families and the dependency family.

Resampling generations reproduces the analytic result. Bootstrapping generations within each condition and length gives a median $|z|$ within 3% of the analytic value, and the Bonferroni verdict ($|z| \geq 3.51$) changes in 1 of 91 cells. The analytic standard errors are therefore not the weak point.

Resampling whole cells is far wider. Treating each (length, spacing, alphabet) cell as the unit instead, so that whole cells enter or leave a resample, widens the median interval from 0.046 to 0.155 and leaves 17 of 91 cells above the Bonferroni threshold instead of 62. Two things drive this, and neither is a defect in the data. First, a pooled cell deliberately spans lengths at which the effect is absent: at $N \leq 16$ these models are near ceiling and markers do nothing, which is itself Finding 2. Averaging those in and then asking how much the average moves when lengths are resampled mostly measures that by-design heterogeneity. Second, a pooled cell contains only 10 to 25 (length, spacing, alphabet) cells, and a bootstrap over so few clusters has little power.

Restricting to the length where the claim is actually made removes the first problem. At $N = 64$, all 24 spaced-letter and digit cells at the four stronger models have a positive effect, between +0.07 and +0.37 (mean +0.24); all 24 clear $|z| \geq 3.51$ analytically and under the generation bootstrap, 22 of 24 have a cell-level $|z| \geq 2$, and 4 of 24 clear 3.51 under the cell-level bootstrap with its 15 or so clusters (`results_clean/_summary/finding2_at_N64.csv`).

We therefore read a pooled z as a statement about the mean effect over the grid we ran, not as evidence that the effect is uniform across lengths and spacings, nor that it would survive a different choice of grid. The claims about structure rest on the length-resolved and position-resolved analyses (Figures 4 and 3), not on the pooled numbers.

Dropping the redundant configurations. At $A = 2$ the three position-local rules coincide and at $M = 2$ `running_sum_local` computes `running_xor` (Section 3). Recomputing all 91 pooled cells without those configurations moves the effect by 0.013 in the median cell and at most 0.054; the effect gets *larger* in 54 of 91 cells, because the two-symbol cells are the easiest and dilute the average. Three cells change sign, all with $|\Delta| \leq 0.009$ and none significant either way, and the Bonferroni verdict changes in 5 of 91 cells. Finding 2 is unchanged: 23 of its 24 cells remain significant.

D Per-model results

Figures 7 and 8 show accuracy by sequence length for each open model and task; Gemini is in Table 14. Pooled effects and their significance are in Figure 5.

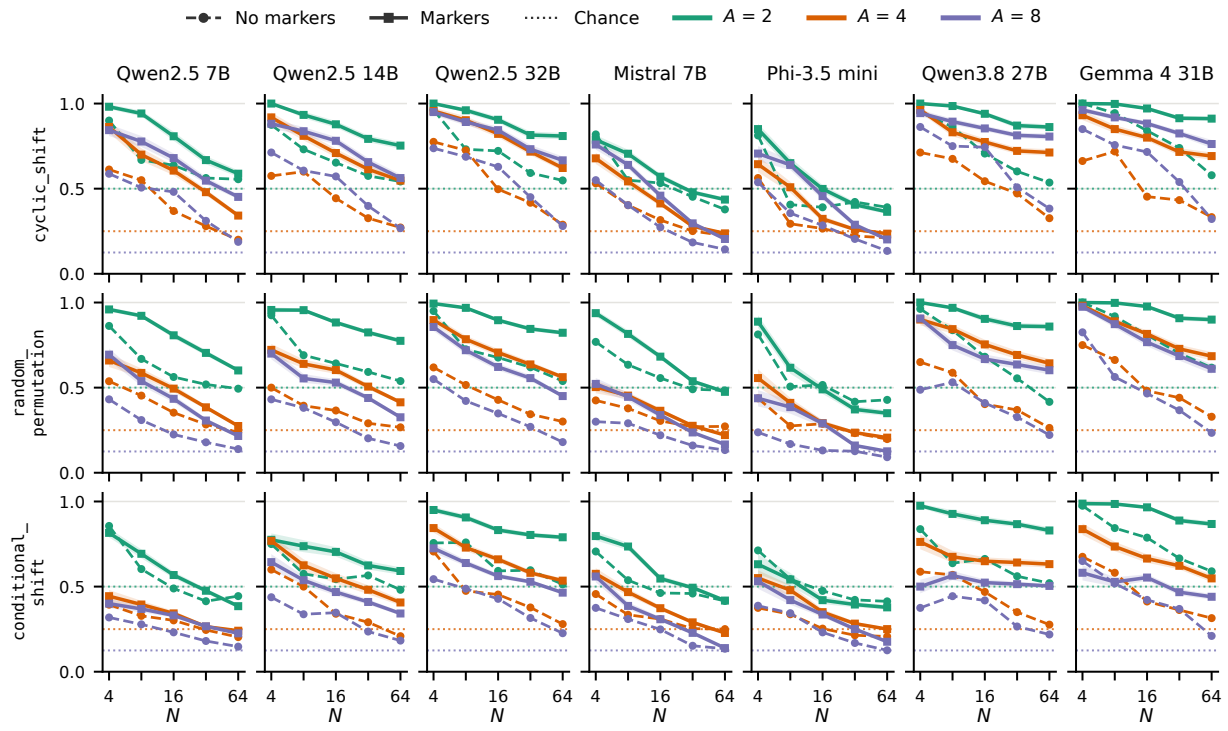


Figure 7: Position-local tasks on contiguous letters, by model and length. Character accuracy against sequence length N (4 to 64). Dashed: no markers; solid: markers (mean over spacings $k \in \{1, 2, 4, 8, 16\}$ with $k < N$); colors: alphabet size; dotted: chance; bands: ±1 standard error.

Mistral-7B. In all three position-local tasks, the benefit is largest at short lengths and decays to zero at the longest length tested (`cyclic_shift`, $N = 256$: +0.008, $z = 0.4$; `random_permutation`, $N = 64$:

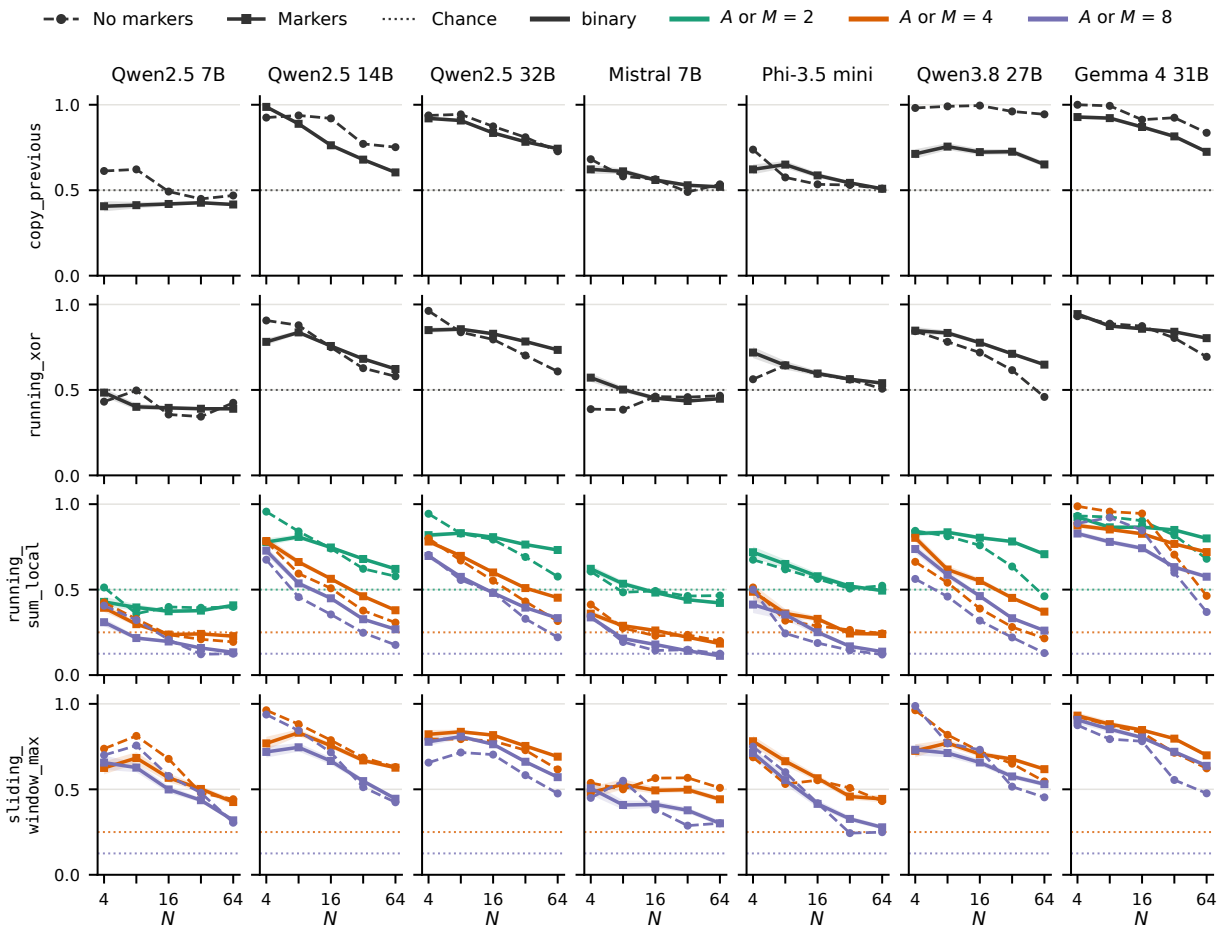


Figure 8: Dependency tasks (digits), by model and length; same conventions as Figure 7. Colors: modulus (`running_sum_local`) or alphabet size (`sliding_window_max`); black: binary. Marker runs average over all positions, so they mix the post-marker loss with the within-segment gain (Section 6).

-0.009, $z = 0.5$; `conditional_shift`, $N = 64$: -0.006, $z = 0.4$). In Qwen models the benefit holds or grows with length.

Repetition collapse. On `running_xor`, Qwen2.5-7B repeats a single wrong digit in about 38% of trials, more often under dense markers. This failure falls to 0.9% at 14B and 0.1% for Mistral-7B, so it is specific to Qwen2.5-7B.

Out-of-alphabet answers. On the digit tasks, Qwen2.5-7B often writes a letter (usually `t`, sometimes `o`) in place of `0`, in 34% to 39% of outputs, with or without markers. On spaced letters, several models step outside the alphabet (for example `I` after `H` instead of wrapping to `A`). Such characters are scored as errors.

E Per-position accuracy

Table 9 gives the per-position means plotted in Figure 3, and Table 10 breaks them down by distance from the marker. Table 11 repeats the dependency-task comparison under alignment-robust scoring, and Figure 9 shows the length profile of the tokenization controls for each task. "No markers" is the no-marker run's accuracy at the same position indices. Means are over the tasks in each block; per-task values are in `results_clean/_summary/per_position_values.csv`.

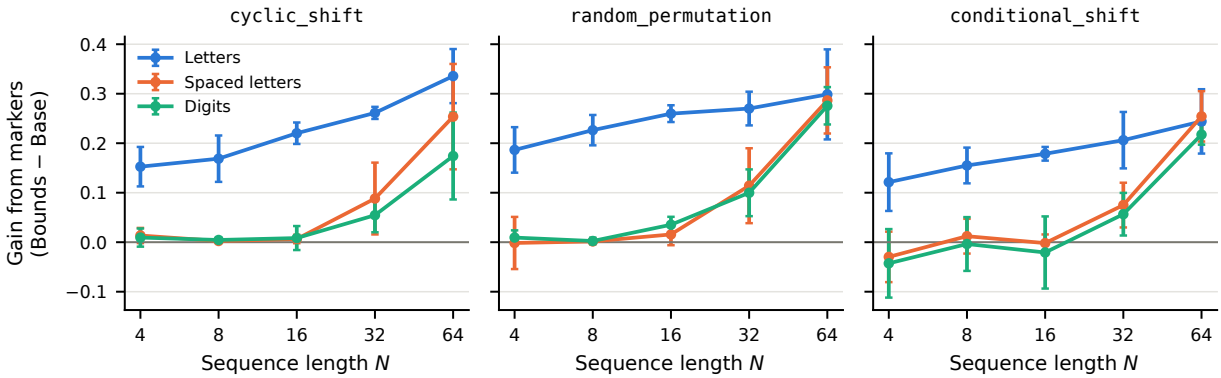


Figure 9: The length profile of Figure 4a, task by task (mean over Qwen2.5-14B and 32B, Qwen3.8-27B, and Gemma-4-31B; whiskers are 95% intervals over models). All three tasks show the same pattern.

Table 9: Per-position accuracy by model ($N \leq 64$, $k \in \{2, 4, 8, 16\}$), mean over the tasks in each block.

	Within a segment			Right after a marker		
Model	No markers	Markers	Δ	No markers	Markers	Δ
Position-local tasks, letters						
Qwen2.5 7B	0.38	0.45	+0.07	0.37	0.51	+0.14
Qwen2.5 14B	0.43	0.55	+0.11	0.40	0.63	+0.23
Qwen2.5 32B	0.49	0.65	+0.16	0.45	0.79	+0.34
Mistral 7B	0.35	0.35	0.00	0.30	0.39	+0.09
Phi-3.5 mini	0.31	0.31	0.00	0.28	0.35	+0.07
Qwen3.8 27B	0.50	0.69	+0.20	0.47	0.81	+0.34
Gemma 4 31B	0.56	0.72	+0.16	0.51	0.80	+0.29
Gemini 3.5 Flash	0.57	0.74	+0.17	0.52	0.83	+0.32
Position-local tasks, spaced letters						
Qwen2.5 7B	0.54	0.55	+0.01	0.52	0.52	0.00
Qwen2.5 14B	0.69	0.80	+0.11	0.67	0.82	+0.15
Qwen2.5 32B	0.79	0.93	+0.14	0.79	0.92	+0.14
Mistral 7B	0.40	0.43	+0.03	0.35	0.45	+0.09
Phi-3.5 mini	0.35	0.35	−0.01	0.33	0.34	+0.01
Qwen3.8 27B	0.83	0.92	+0.09	0.84	0.92	+0.08
Gemma 4 31B	0.88	0.94	+0.06	0.89	0.94	+0.05
Position-local tasks, digits						
Qwen2.5 7B	0.35	0.38	+0.04	0.30	0.35	+0.05
Qwen2.5 14B	0.77	0.85	+0.08	0.77	0.87	+0.09
Qwen2.5 32B	0.83	0.94	+0.11	0.83	0.93	+0.09
Mistral 7B	0.43	0.38	−0.04	0.41	0.39	−0.02
Phi-3.5 mini	0.48	0.55	+0.07	0.44	0.56	+0.12
Qwen3.8 27B	0.85	0.94	+0.09	0.85	0.94	+0.09
Gemma 4 31B	0.92	0.99	+0.07	0.92	0.98	+0.06
Dependency tasks						
Qwen2.5 7B	0.43	0.41	−0.02	0.42	0.36	−0.06
Qwen2.5 14B	0.68	0.68	0.00	0.64	0.49	−0.15
Qwen2.5 32B	0.70	0.77	+0.07	0.66	0.52	−0.15
Mistral 7B	0.44	0.43	0.00	0.43	0.38	−0.04
Phi-3.5 mini	0.49	0.48	0.00	0.45	0.42	−0.03
Qwen3.8 27B	0.68	0.70	+0.02	0.63	0.44	−0.19
Gemma 4 31B	0.79	0.85	+0.05	0.78	0.52	−0.26
Gemini 3.5 Flash	0.92	0.97	+0.05	0.92	0.49	−0.43
Dependency tasks, reset variants						
Qwen2.5 7B	0.43	0.43	0.00	0.41	0.58	+0.17
Qwen2.5 32B	0.69	0.79	+0.10	0.65	0.92	+0.27
Qwen3.8 27B	0.66	0.68	+0.03	0.63	0.98	+0.36
Gemma 4 31B	0.77	0.85	+0.08	0.75	0.97	+0.22

Table 10: Accuracy by distance after the marker at $k = 16$ ($N \in \{32, 64\}$): 0 is the position right after the marker. Mean over the tasks in each block.

		Positions after the marker				
Tasks	Model	0	1	2–3	4–7	8–15
Position-local, letters	Qwen2.5 7B	0.32	0.38	0.35	0.29	0.26
	Qwen2.5 14B	0.45	0.47	0.41	0.37	0.34
	Qwen2.5 32B	0.60	0.57	0.48	0.45	0.41
	Mistral 7B	0.28	0.34	0.29	0.26	0.25
	Phi-3.5 mini	0.30	0.28	0.28	0.28	0.23
	Qwen3.8 27B	0.68	0.67	0.59	0.51	0.47
	Gemma 4 31B	0.52	0.62	0.55	0.51	0.43
	Gemini 3.5 Flash	0.74	0.69	0.65	0.51	0.46
Position-local, spaced	Qwen2.5 7B	0.30	0.46	0.47	0.43	0.35
	Qwen2.5 14B	0.59	0.67	0.68	0.66	0.59
	Qwen2.5 32B	0.86	0.90	0.89	0.88	0.85
	Mistral 7B	0.31	0.39	0.37	0.27	0.24
	Phi-3.5 mini	0.25	0.37	0.33	0.25	0.21
	Qwen3.8 27B	0.89	0.88	0.91	0.91	0.90
	Gemma 4 31B	0.93	0.94	0.95	0.93	0.94
Position-local, digits	Qwen2.5 7B	0.28	0.38	0.32	0.30	0.25
	Qwen2.5 14B	0.83	0.85	0.86	0.81	0.76
	Qwen2.5 32B	0.90	0.93	0.94	0.93	0.90
	Mistral 7B	0.31	0.38	0.36	0.31	0.26
	Phi-3.5 mini	0.38	0.53	0.52	0.46	0.37
	Qwen3.8 27B	0.90	0.89	0.91	0.91	0.90
	Gemma 4 31B	0.94	0.98	0.99	0.98	0.98
Dependency	Qwen2.5 7B	0.36	0.41	0.42	0.42	0.36
	Qwen2.5 14B	0.45	0.75	0.68	0.60	0.53
	Qwen2.5 32B	0.49	0.85	0.79	0.71	0.63
	Mistral 7B	0.41	0.42	0.45	0.42	0.41
	Phi-3.5 mini	0.40	0.47	0.44	0.40	0.38
	Qwen3.8 27B	0.46	0.70	0.71	0.65	0.58
	Gemma 4 31B	0.49	0.82	0.83	0.81	0.75
	Gemini 3.5 Flash	0.57	1.00	1.00	1.00	0.97

Table 11: Per-position gains on dependency tasks (marker run minus the no-marker run at the same positions) under index-by-index scoring and under edit-distance alignment, which repairs dropped or inserted characters. $N \leq 64$, $k \in \{2, 4, 8, 16\}$.

		Right after a marker		Within a segment	
Task	Model	Index	Aligned	Index	Aligned
`copy_previous`	Qwen2.5 7B	−0.10	−0.09	−0.07	−0.06
	Qwen2.5 14B	−0.21	−0.15	−0.10	−0.05
	Qwen2.5 32B	−0.15	−0.17	−0.02	−0.02
	Mistral 7B	−0.05	−0.01	0.00	+0.01
	Phi-3.5 mini	−0.04	−0.07	−0.01	−0.04
	Qwen3.8 27B	−0.45	−0.20	−0.24	−0.05
	Gemma 4 31B	−0.29	−0.25	−0.07	−0.04
`running_xor`	Qwen2.5 7B	+0.02	−0.02	+0.01	0.00
	Qwen2.5 14B	−0.14	−0.13	+0.05	+0.01
	Qwen2.5 32B	−0.19	−0.24	+0.11	+0.03
	Mistral 7B	−0.04	−0.03	+0.02	−0.03
	Phi-3.5 mini	−0.05	−0.02	+0.03	+0.03
	Qwen3.8 27B	−0.06	−0.08	+0.13	+0.06
	Gemma 4 31B	−0.26	−0.29	+0.09	+0.04
`running_sum_local`	Qwen2.5 7B	−0.03	−0.04	−0.01	−0.02
	Qwen2.5 14B	−0.08	−0.12	+0.07	+0.03
	Qwen2.5 32B	−0.16	−0.22	+0.11	+0.05
	Mistral 7B	−0.01	−0.04	0.00	−0.01
	Phi-3.5 mini	−0.03	0.00	−0.02	−0.03
	Qwen3.8 27B	−0.06	−0.13	+0.14	+0.09
	Gemma 4 31B	−0.41	−0.46	+0.11	+0.07
`sliding_window_max`	Qwen2.5 7B	−0.11	−0.06	−0.02	−0.02
	Qwen2.5 14B	−0.16	−0.10	−0.03	−0.01
	Qwen2.5 32B	−0.09	−0.07	+0.08	+0.07
	Mistral 7B	−0.07	0.00	−0.02	0.00
	Phi-3.5 mini	+0.02	−0.03	−0.02	−0.03
	Qwen3.8 27B	−0.18	−0.16	+0.05	+0.04
	Gemma 4 31B	−0.07	−0.08	+0.08	+0.02

F Reset variants, pooled

Table 12 gives the pooled effect of markers on each dependency task in its original and reset forms; the per-position comparison is in Section 6.1.

Table 12: Reset variants, pooled over positions: Character accuracy Base → Bounds, Δ, and $|z|$, pooled as in Figure 5 (no-marker mean length-matched). Base differs between the original and reset columns because the reset prompt alone shifts no-marker accuracy; Section 6.1 therefore compares marker runs position by position. Originals use 20 to 80 trials per cell (Table 8); reset variants 20, except 80 at Qwen2.5-7B.

		Original (dependency intact)		Reset (dependency severed)	
Task	Model	Base→Bounds	Δ ($\|z\|$)	Base→Bounds	Δ ($\|z\|$)
`running_xor`	7B	0.401→0.403	+0.002 (0.1)	0.450→0.482	+0.032 (3.4)
	32B	0.746→0.798	+0.053 (6.2)	0.715→0.878	+0.163 (11.9)
	Qwen3.8 27B	0.646→0.742	+0.096 (9.5)	0.645→0.855	+0.210 (14.4)
	Gemma 4 31B	0.816→0.850	+0.034 (4.3)	0.816→0.928	+0.112 (10.2)
`running_sum_local`	7B	0.286→0.280	−0.006 (0.8)	0.277→0.348	+0.071 (12.4)
	32B	0.547→0.603	+0.056 (9.0)	0.508→0.660	+0.152 (12.1)
	Qwen3.8 27B	0.441→0.573	+0.131 (13.0)	0.419→0.683	+0.264 (19.4)
	Gemma 4 31B	0.754→0.775	+0.021 (3.2)	0.723→0.840	+0.117 (10.8)
`sliding_window_max`	32B	0.668→0.733	+0.065 (7.9)	0.677→0.816	+0.139 (10.9)
	Qwen3.8 27B	0.666→0.653	−0.013 (1.0)	0.640→0.824	+0.183 (16.4)
	Gemma 4 31B	0.708→0.783	+0.075 (6.3)	0.718→0.817	+0.099 (7.9)
`copy_previous`	7B	0.508→0.418	−0.090 (7.0)	0.565→0.553	−0.013 (1.4)
	32B	0.836→0.818	−0.018 (1.9)	0.874→0.792	−0.083 (4.6)
	Qwen3.8 27B	0.971→0.709	−0.262 (23.9)	0.942→0.836	−0.105 (6.4)
	Gemma 4 31B	0.917→0.832	−0.086 (6.5)	0.837→0.866	+0.029 (1.3)

G External tasks

Tables 13 and 14 and Figures 10 and 11 give the results on tasks we did not design (Section 6) and on Gemini.

Table 13: External position-local tasks, pooled across length and density. Base: no-marker accuracy averaged over lengths. Best-k: accuracy at the single best spacing (optimistic). Δ: marker runs minus the length-matched no-marker mean (Section 3). ProcBench-Substitute was regenerated after we made the vendored ProcBench generator's substitution table deterministic (it was drawn in a process-dependent order).

	Qwen2.5-7B			Qwen2.5-32B		
Task	Base	Best-k	Δ ($\|z\|$)	Base	Best-k	Δ ($\|z\|$)
Vowel/consonant tagging	0.438	0.435	−0.083 (2.4)	0.615	0.657	+0.012 (0.5)
ProcBench-Substitute	0.189	0.432	+0.096 (6.7)	0.390	0.757	+0.203 (10.8)
String copying	0.973	1.000	+0.009 (1.0)	0.993	1.000	+0.007 (2.0)

On Kadane's algorithm, base accuracy at both Qwen scales is statistically indistinguishable from chance under a shuffle test, so we do not interpret its effects. Minimum is well above chance at Qwen2.5-32B ($z = 4.1$ from shuffled chance).

Table 14: Closed-model check and CLRS-Text. Gemini 3.5 Flash with 3 trials per cell, pooled over all tested lengths (including N up to 256 for `cyclic_shift`), with the no-marker mean length-matched. CLRS-Text tasks use token-level markers and scoring.

Task	Model	Base acc.	Bounds acc.	Δ ($\lvert z \rvert$)
`cyclic_shift`	Gemini	0.596	0.852	+0.256 (14.4)
`random_permutation`	Gemini	0.561	0.846	+0.284 (8.4)
`conditional_shift`	Gemini	0.453	0.628	+0.175 (3.1)
`running_xor`	Gemini	0.882	0.913	+0.032 (0.4)
`running_sum_local`	Gemini	0.918	0.803	−0.116 (5.2)
`copy_previous`	Gemini	0.984	0.924	−0.060 (2.8)
CLRS minimum	Qwen2.5-7B	0.133	0.183	+0.050 (3.3)
	Qwen2.5-32B	0.592	0.392	−0.200 (4.5)
	Gemini	0.997	0.654	−0.343 (7.3)
CLRS Kadane	Qwen2.5-7B	0.060	0.068	+0.008 (1.0)
	Qwen2.5-32B	0.186	0.122	−0.064 (2.0)
	Gemini	0.384	0.354	−0.030 (0.3)

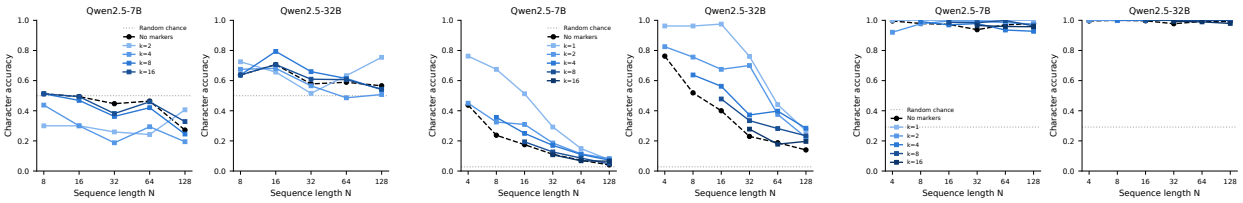


Figure 10: External tasks by marker density. Left: vowel/consonant tagging. Middle: ProcBench-Substitute. Right: string copying.

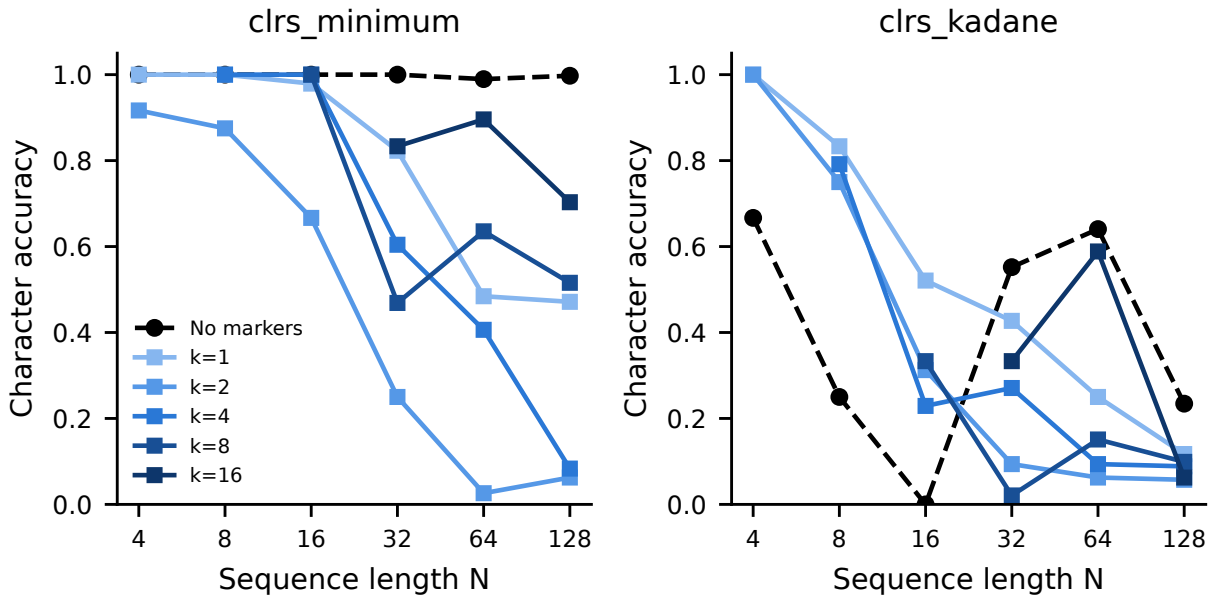


Figure 11: CLRS-Text tasks on Gemini 3.5 Flash by sequence length, one line per marker spacing k (3 trials per cell). Dashed: no markers.

H Mechanism details

This appendix collects the attention analyses behind Section 7: the mechanism overview (Figure 12), attention statistics by layer (Figure 13), example heads (Figure 14), the earlier head ablation (Figure 15), a digit-tokenization check of the attention pattern (Figure 16), error recovery after markers (Figure 17), the head taxonomy (Table 16), and the 7B ablation (Table 15). In Figure 17, the 50% rate away from markers is chance for this binary alphabet.

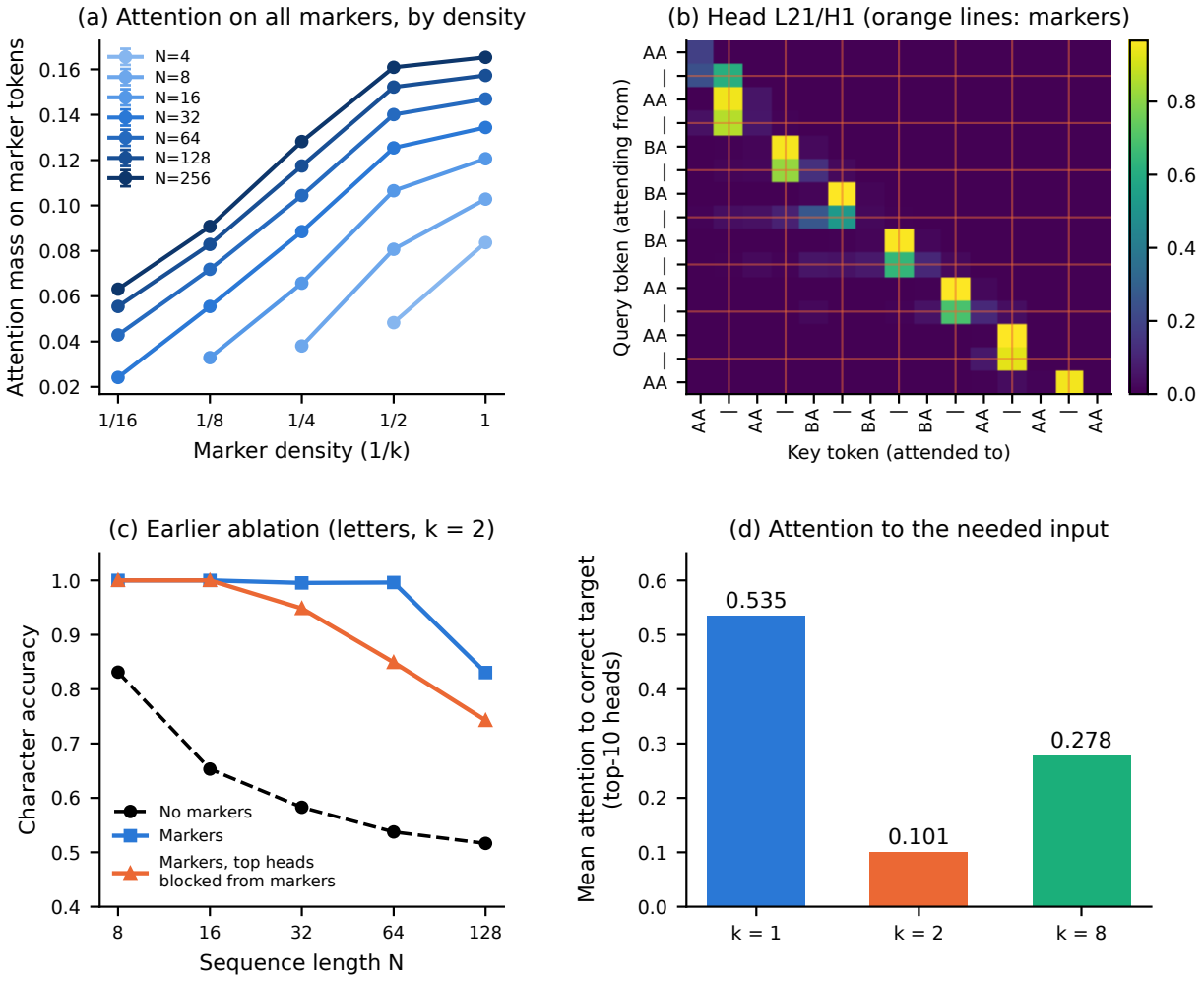


Figure 12: Mechanism (Qwen2.5-32B). (a) Total attention mass on all marker tokens rises with density at every length (`cyclic_shift`, $A = 2$); the number of markers rises faster, so this is not evidence that individual markers draw more attention than content tokens. (b) The most marker-attending head (layer 21, head 1) attends from each position to an adjacent marker. (c) Earlier ablation on contiguous letters ($k = 2$; Section 7): masking the ten most marker-attending heads' attention to markers removes a growing share of the benefit as N grows (0% at $N \leq 16$, 11% at 32, 32% at 64, 28% at 128). (d) `copy_previous` at $N = 32$: mean attention from each output position to the input token it needs (top-10 heads) is 0.535 at $k = 1$, 0.101 at $k = 2$, and 0.278 at $k = 8$, matching accuracy (Base 0.83; $k = 1$: 0.88; $k = 2$: 0.65; $k = 8$: 0.84).

Table 15: Head ablation at Qwen2.5-7B, same design as Table 3. Marker gains are small at this scale, so shares are noisy; "–" marks cells with a gain below 0.03.

				Markers, with heads blocked			Share of gain removed	
Format	N	No markers	Markers	Top 10	Random 10	Top 10, sham	Top 10	Controls
Letters	32	0.17	0.24	0.23	0.24	0.24	14%	0%
Letters	64	0.14	0.16	0.14	0.15	0.16	–	–
Spaced letters	32	0.42	0.55	0.51	0.52	0.55	26%	$\leq$ 22%
Spaced letters	64	0.19	0.30	0.27	0.31	0.32	24%	0%
Digits	32	0.24	0.27	0.28	0.29	0.28	–	–
Digits	64	0.18	0.19	0.19	0.19	0.19	–	–

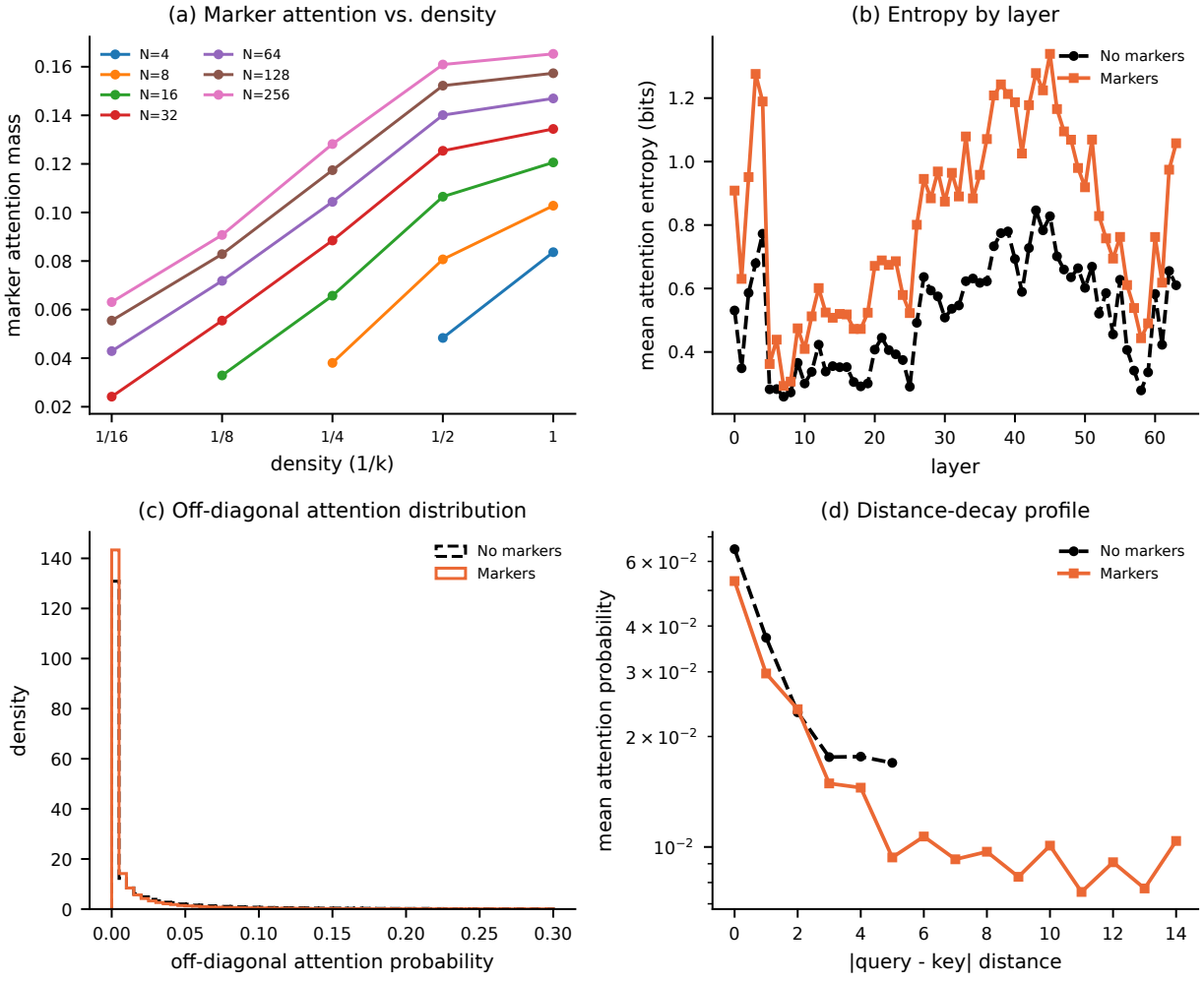


Figure 13: Attention analysis, Qwen2.5-32B, `cyclic_shift`: marker attention by density, attention entropy by layer (higher under markers at nearly every layer), off-diagonal attention, and distance decay.

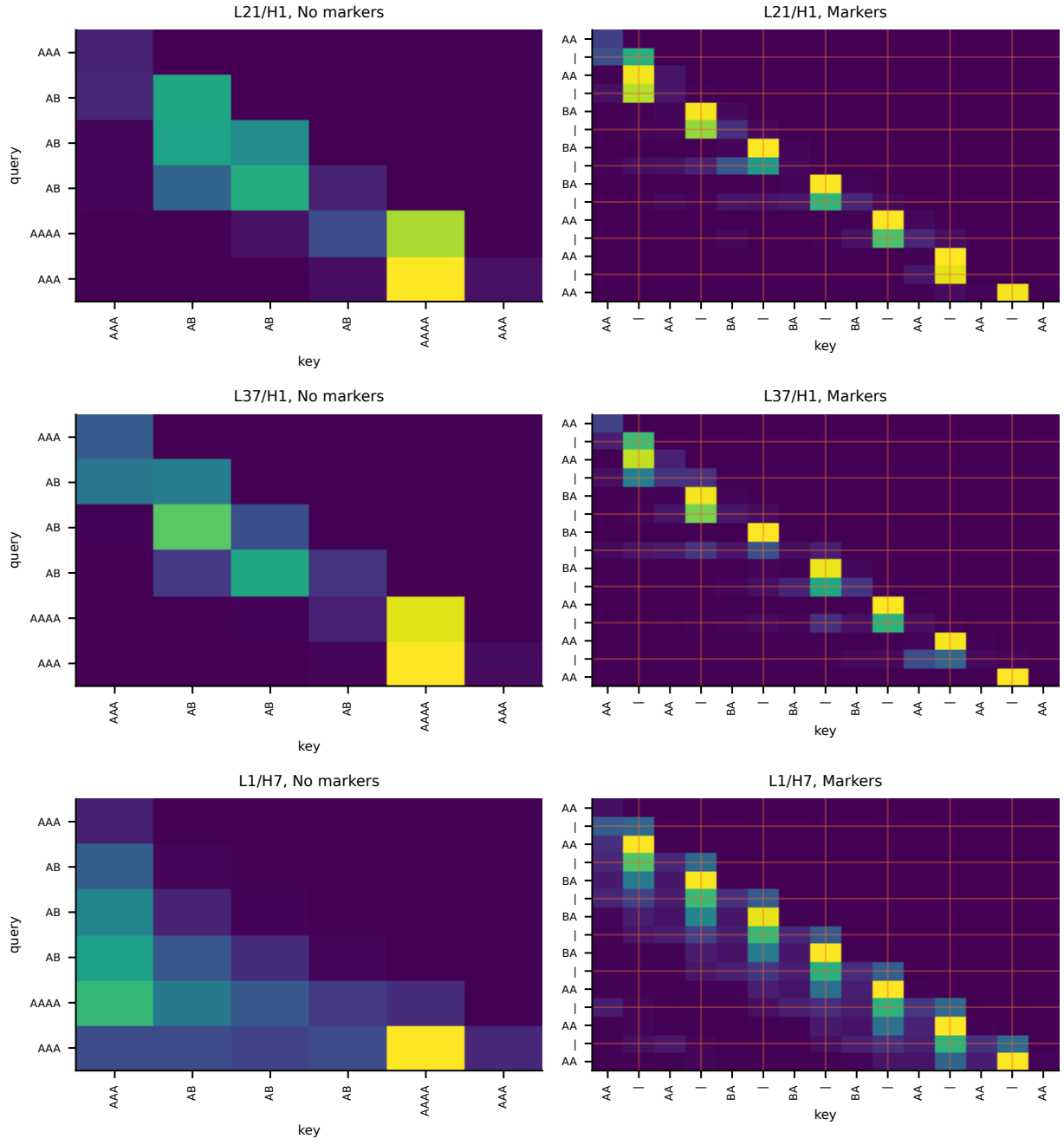


Figure 14: The three most marker-attending heads at Qwen2.5-32B (L21/H1, L37/H1, L1/H7). Without markers, letters merge into multi-character tokens; markers restore per-character resolution.

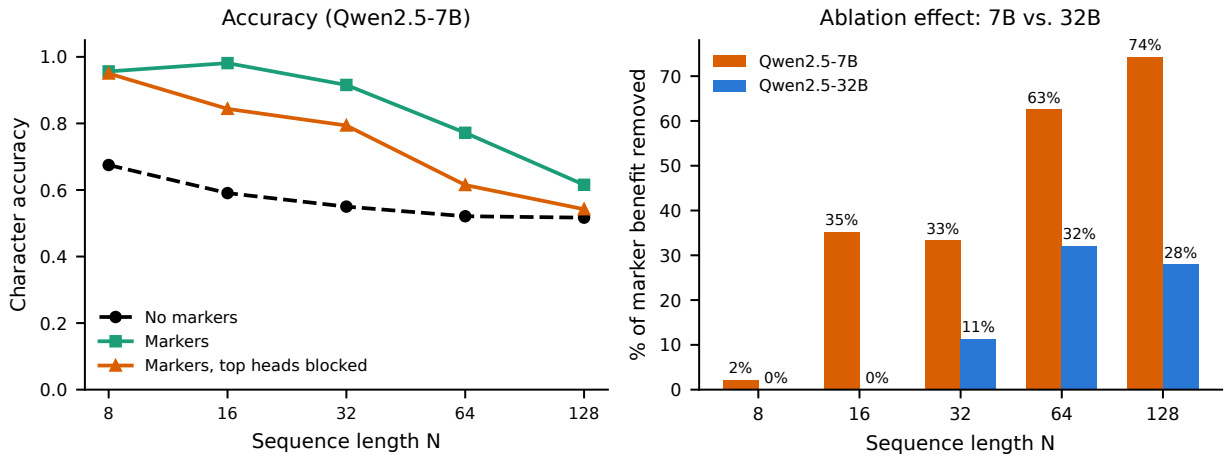


Figure 15: Earlier head ablation, contiguous letters (`cyclic_shift`, $A = 2$, $k = 2$), Qwen2.5-7B and 32B: share of the marker benefit removed by masking the ten most marker-attending heads' attention to markers, re-identified at each length. Unlike Table 3, Base and unablated runs come from an earlier grid and there are no controls at 32B.

Table 16: The ten most marker-attending heads at Qwen2.5-32B, with percentiles within all heads on an induction test and a positional (previous-token) test.

Layer/Head	Marker attention	Induction pct.	Positional pct.
21/1	0.099	38.3	98.3
37/1	0.087	47.4	96.7
1/7	0.086	17.1	97.7
35/19	0.082	23.1	96.5
35/3	0.080	14.7	97.2
30/32	0.079	18.6	96.0
32/37	0.075	14.3	98.8
1/8	0.072	64.8	97.7
32/7	0.071	60.4	90.2
19/19	0.068	9.9	83.9
Mean		30.9	95.3

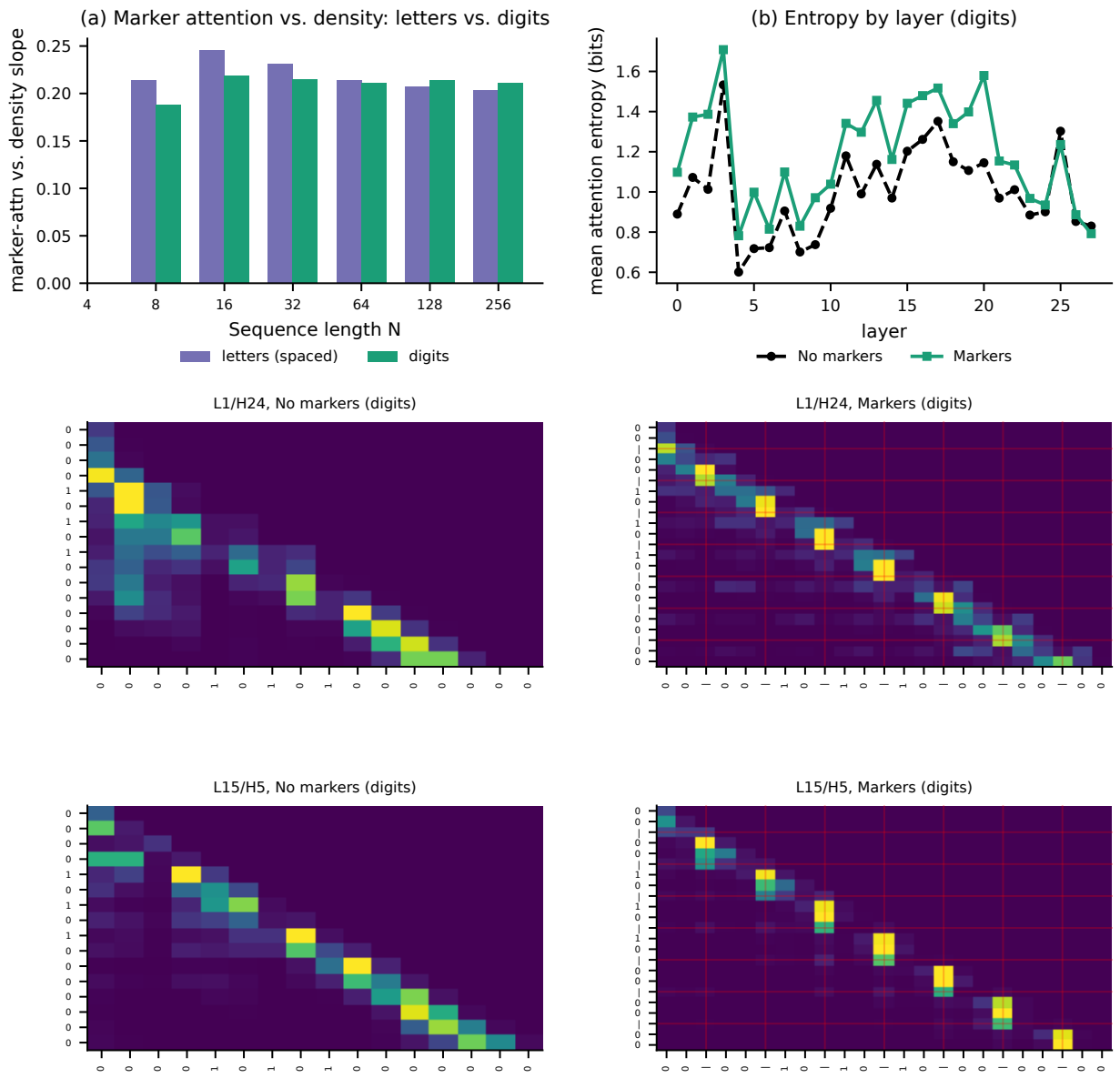


Figure 16: Tokenization check at Qwen2.5-7B: marker-attention slopes with letters (space-separated, one token per letter) and digits (natively one token per character) are nearly identical.

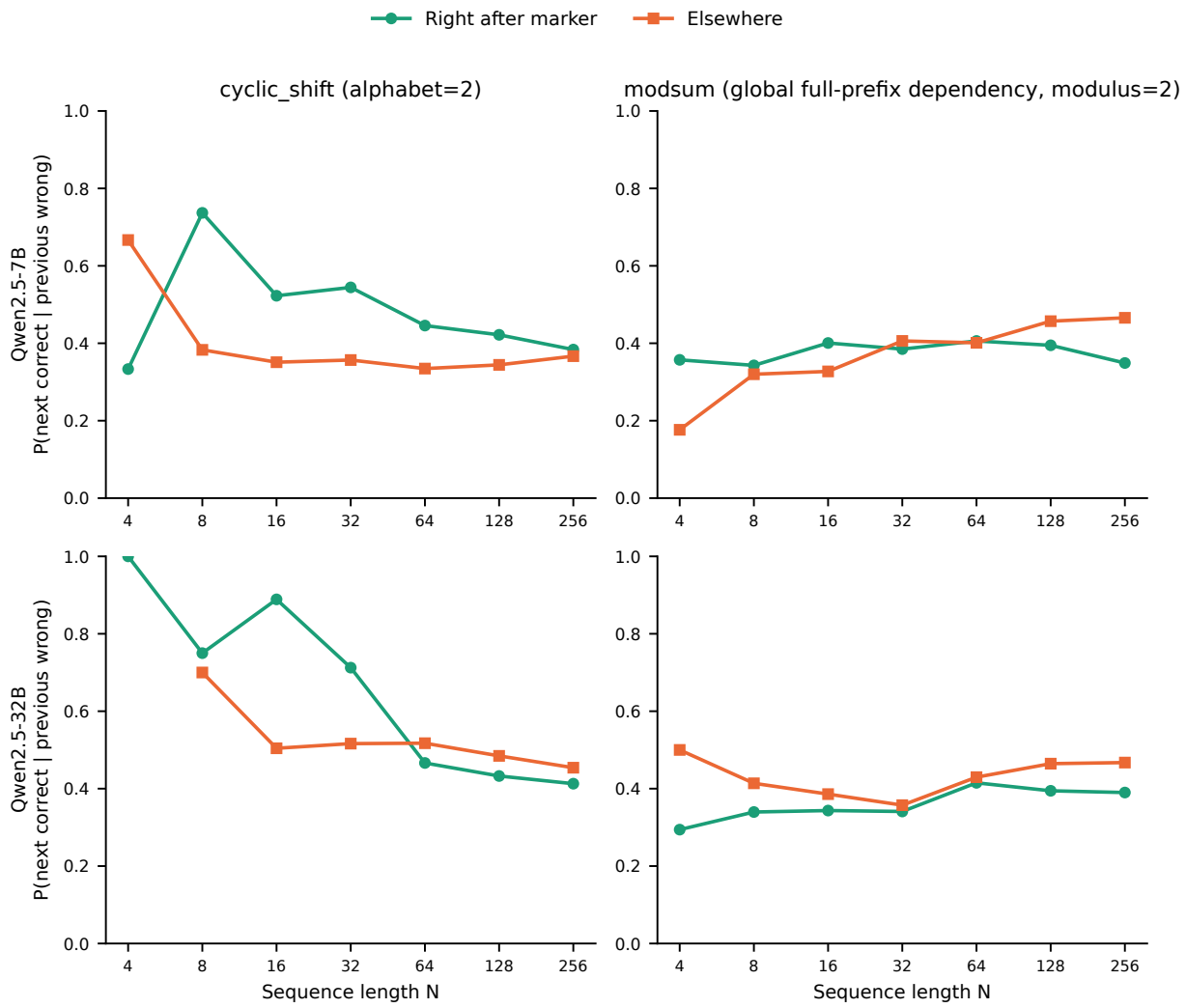


Figure 17: Error recovery: after a wrong output, the fraction of next positions that are correct, split by whether the next position follows a marker. Left: `cyclic_shift`; right: a full-prefix modular sum. $A = M = 2$.